Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemD-apcRev-Da.sil etc.

Creation date: 15-Oct-81 15:26:01

For: Spruce

25 total sheets = 24 pages, 1 copy.

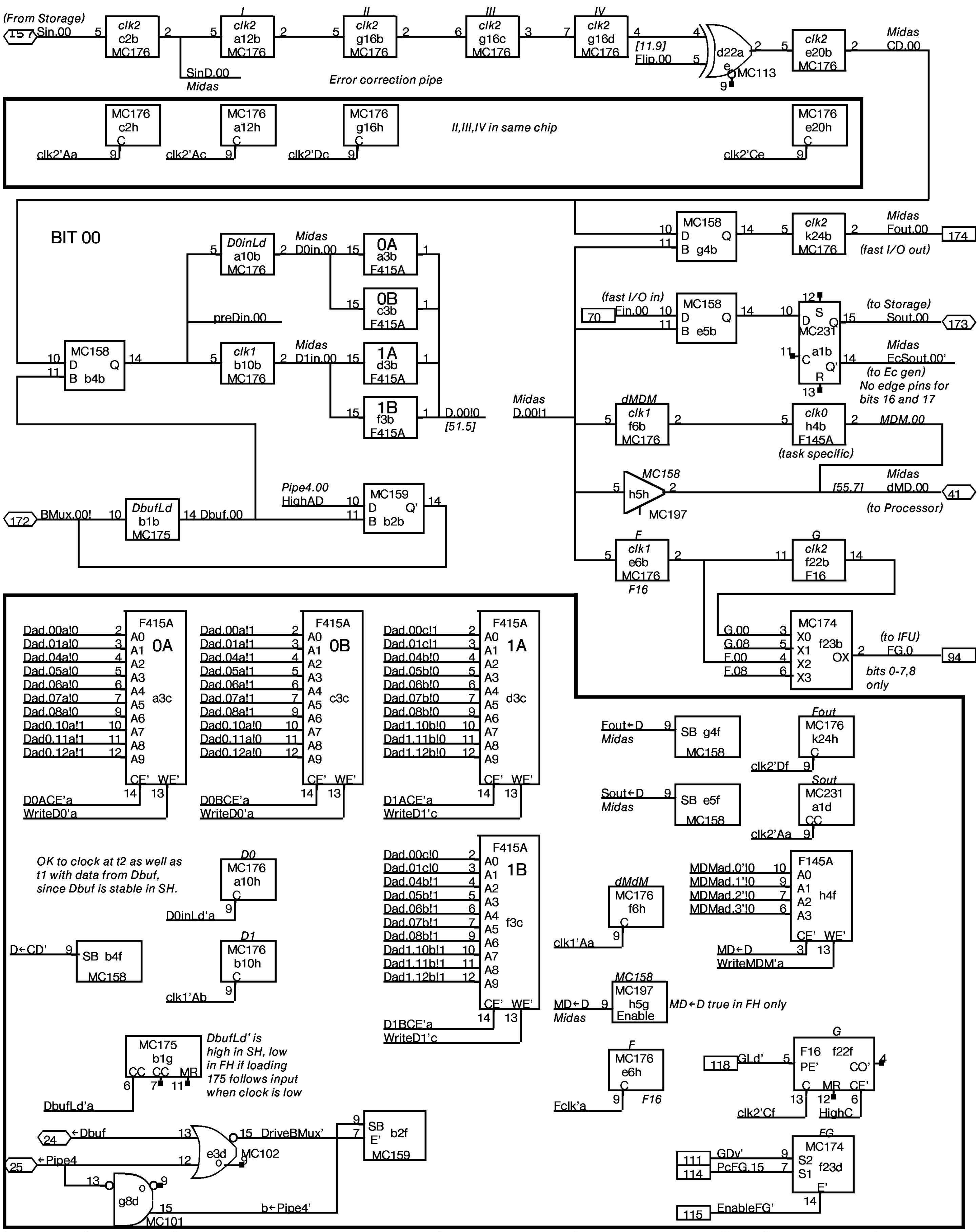
Problems encountered:
Font HELVETICA24B substituted for font HELVE24.

DORADO SCHEMATICS

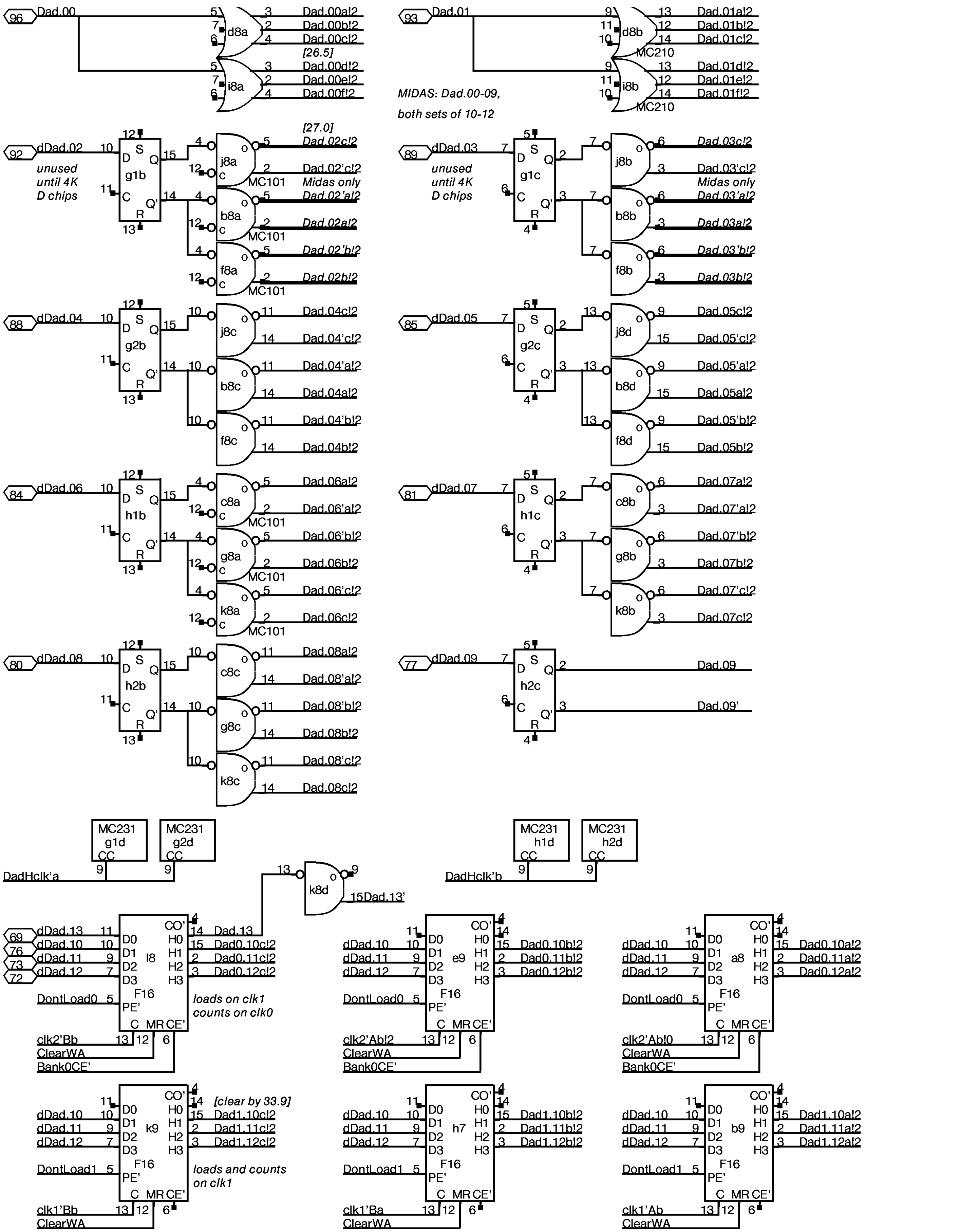
Memory Data Board

(1k chip version)

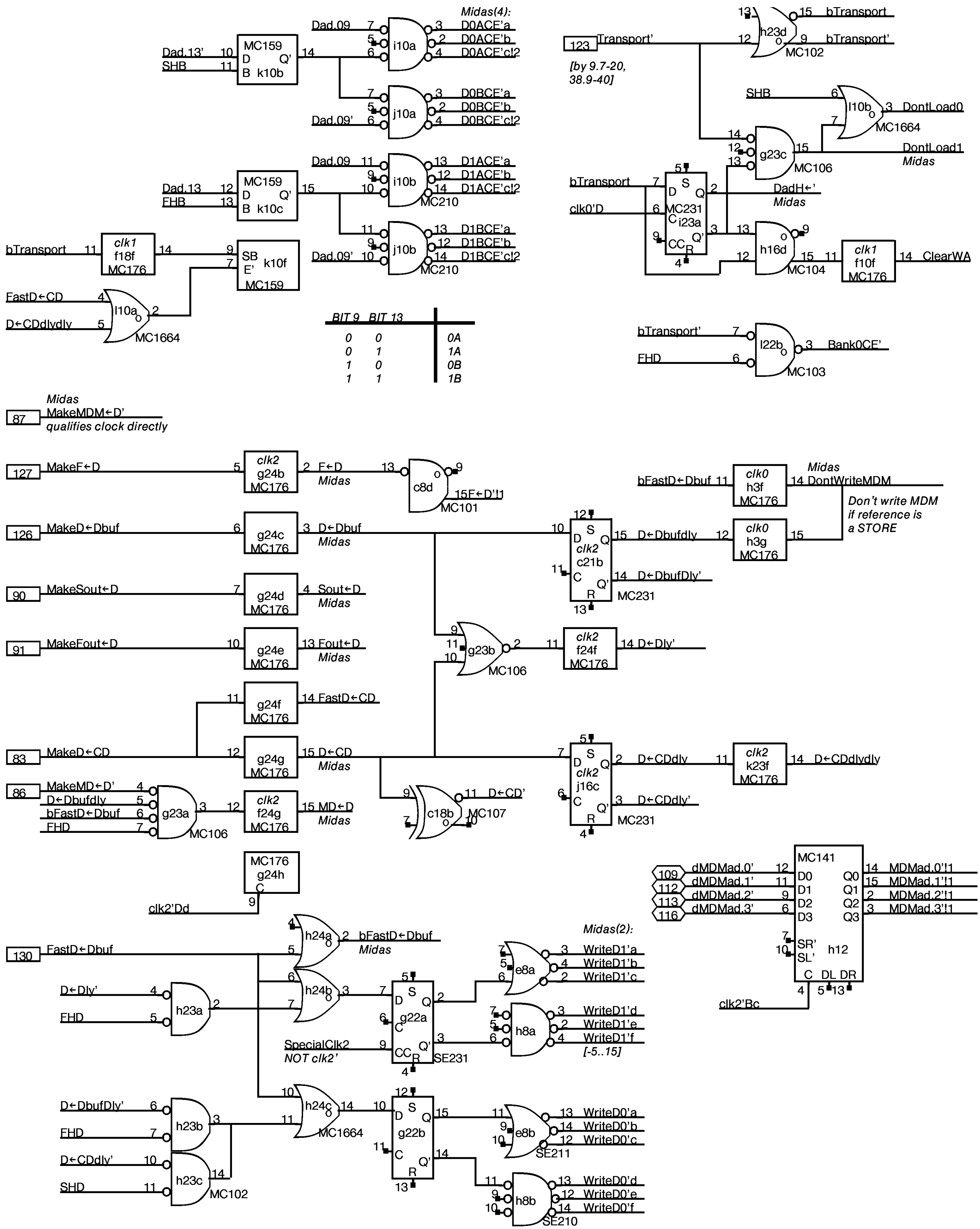
Table of contents



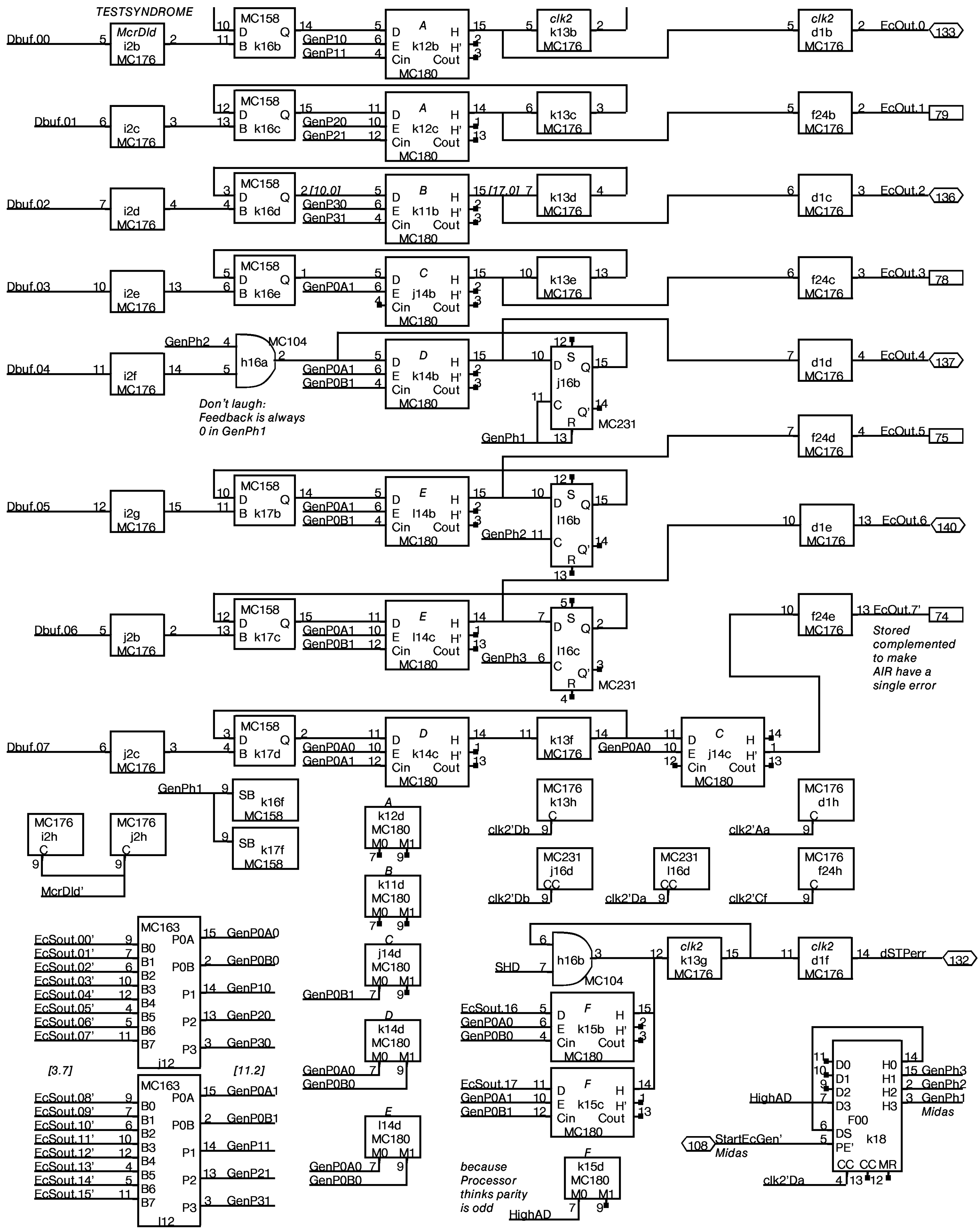
XEROX PARC/CSL	Project Dorado	Memory data board Bit slice 00 + its control	File MemD01.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 01



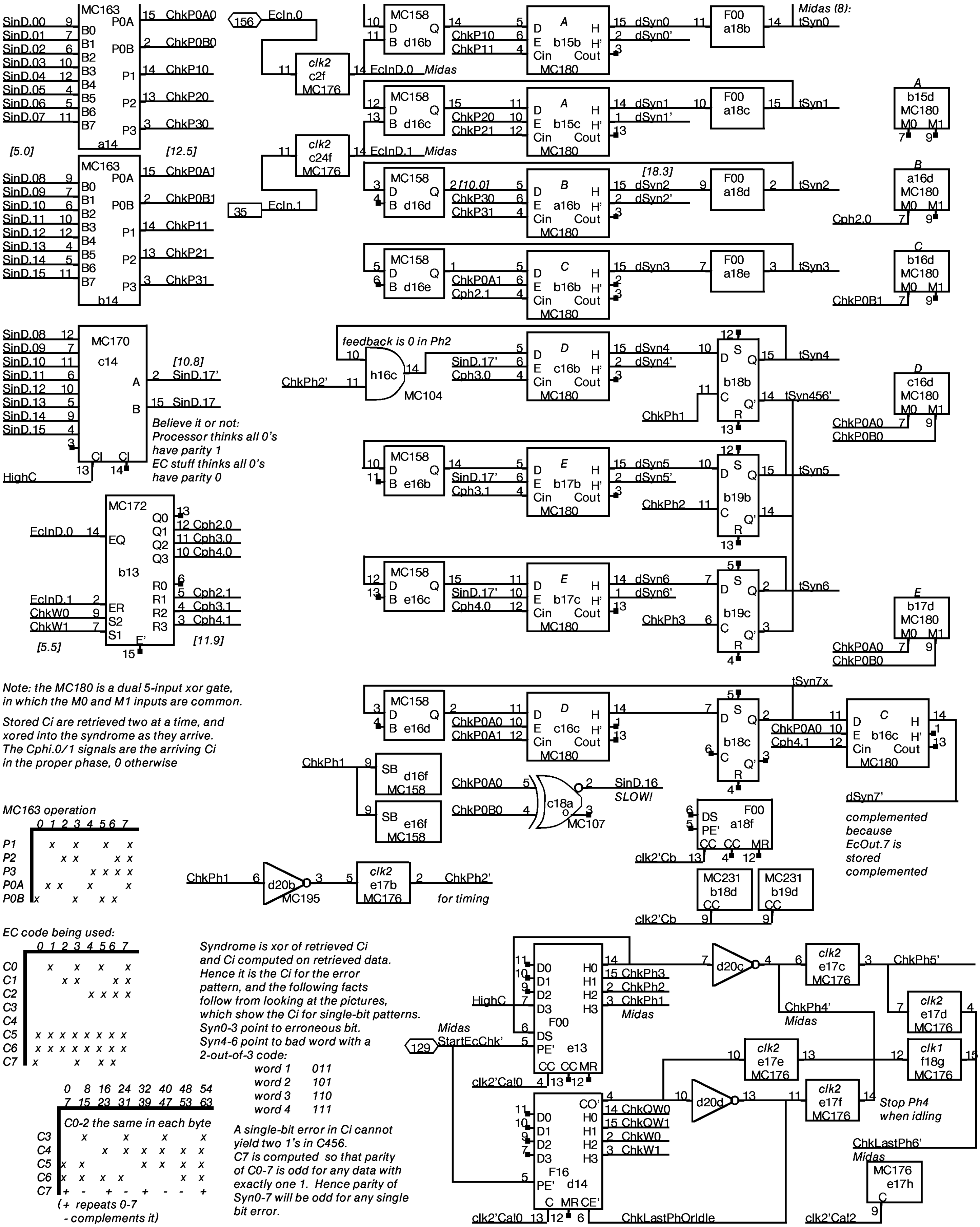
XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board D memory addresses	MemD02.sil	D. Clark	Da	7/09/79	02



XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Control, WE's & CE's	MemD03.sil	D. Clark	Da	7/31/79	03



XEROX PARC/CSL	Project: Dorado	Memory data board Hamming code generator	File: MemD04.sil	Designer: D. Clark	Rev: Da	Date: 7/31/79	Page: 04



Note: the MC180 is a dual 5-input xor gate, in which the M0 and M1 inputs are common.

Stored Ci are retrieved two at a time, and xored into the syndrome as they arrive. The Cphi.0/1 signals are the arriving Ci in the proper phase, 0 otherwise

MC163 operation

	0	1	2	3	4	5	6	7
P1		x		x		x		x
P2			x	x			x	x
P3					x	x	x	x
P0A		x	x		x			x
P0B	x			x		x	x	

EC code being used:

	0	1	2	3	4	5	6	7
C0		x		x		x		x
C1			x	x			x	x
C2					x	x	x	x
C3								
C4								
C5	x	x	x	x	x	x	x	x
C6	x	x	x	x	x	x	x	x
C7	x			x		x	x	

C0-2 the same in each byte

	0 7	8 15	16 23	24 31	32 39	40 47	48 53	54 63
C3		x		x		x		x
C4			x	x	x	x	x	x
C5	x	x			x	x	x	x
C6	x	x	x	x			x	x
C7	+	-	+	-	+	-	-	+

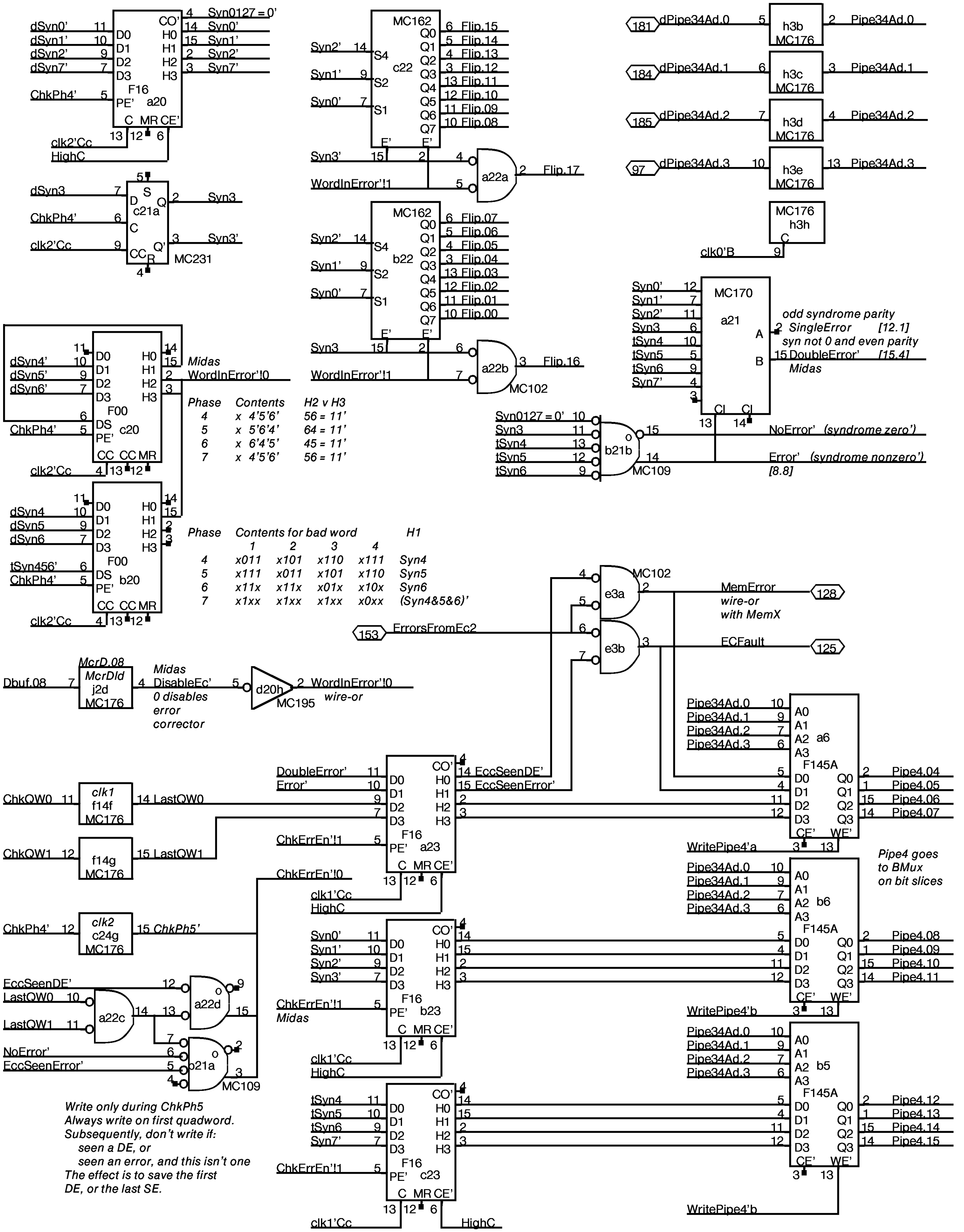
(+ repeats 0-7
- complements it)

Syndrome is xor of retrieved Ci and Ci computed on retrieved data. Hence it is the Ci for the error pattern, and the following facts follow from looking at the pictures, which show the Ci for single-bit patterns. Syn0-3 point to erroneous bit. Syn4-6 point to bad word with a 2-out-of-3 code:

word 1	011
word 2	101
word 3	110
word 4	111

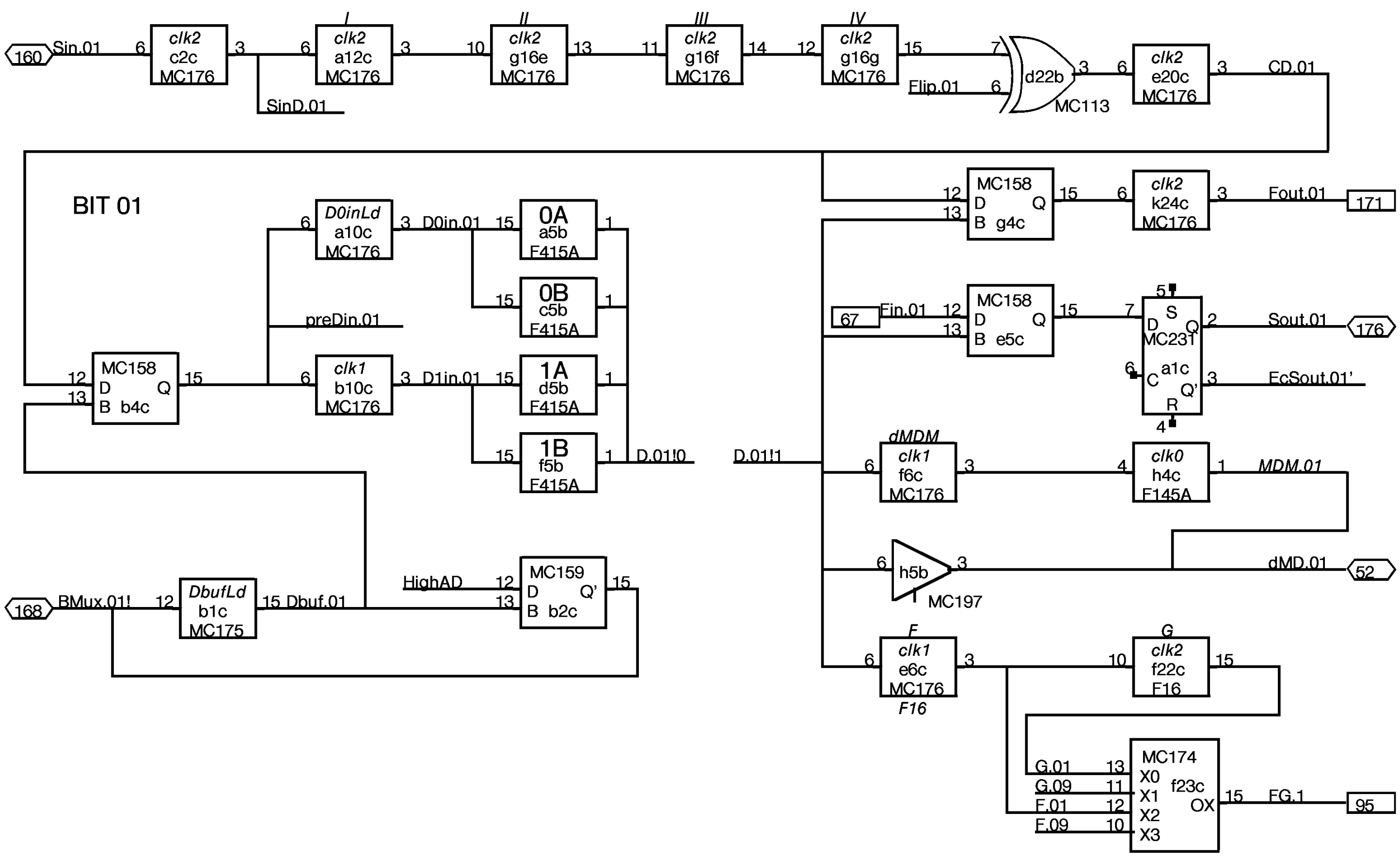
A single-bit error in Ci cannot yield two 1's in C456. C7 is computed so that parity of C0-7 is odd for any data with exactly one 1. Hence parity of Syn0-7 will be odd for any single bit error.

XEROX PARC/CSL	Project Dorado	Memory data board Hamming code checker	File MemD05.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 05

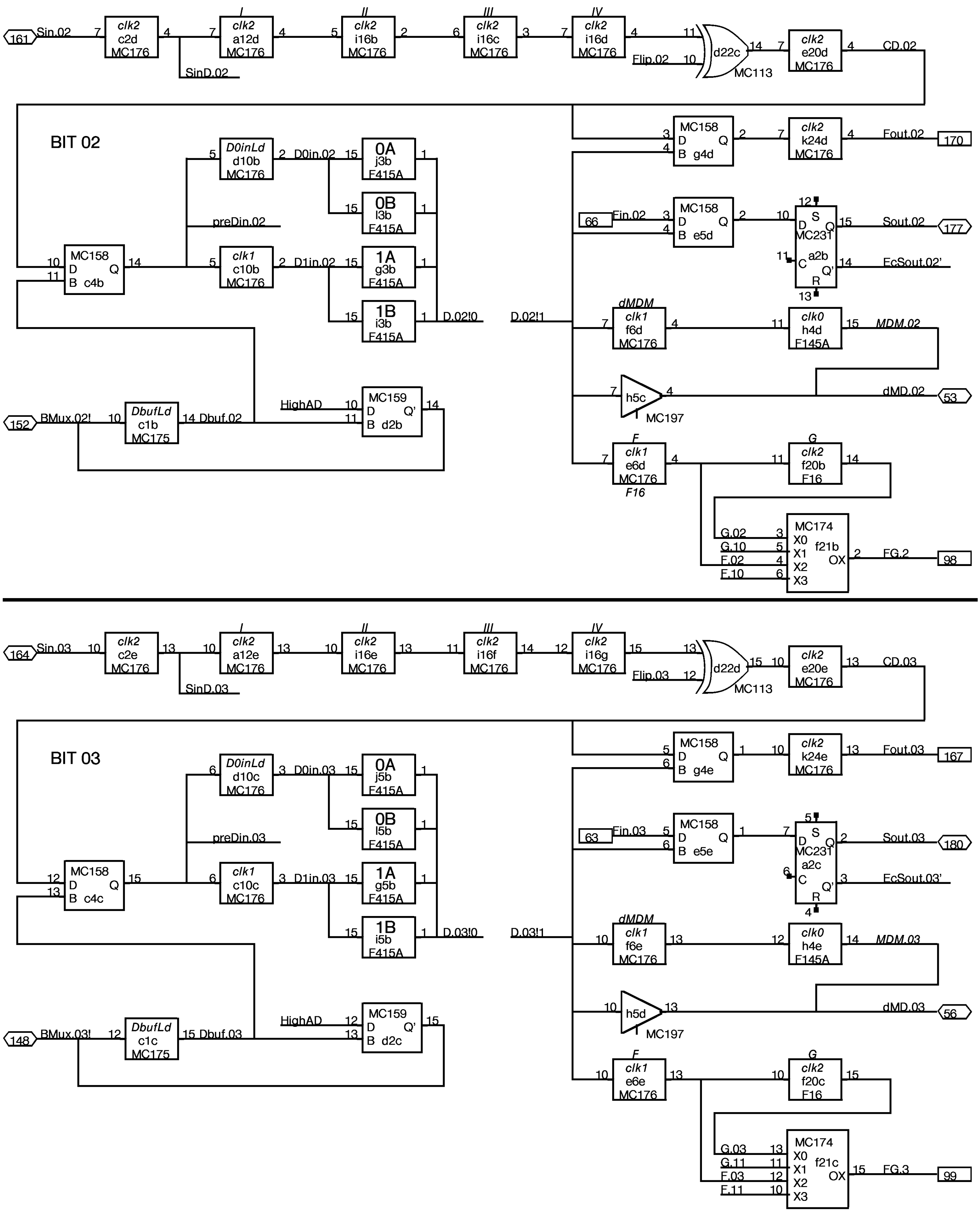


XEROX PARC/CSL	Project Dorado	Memory data board Error corrector, Pipe4	File MemD06.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 06

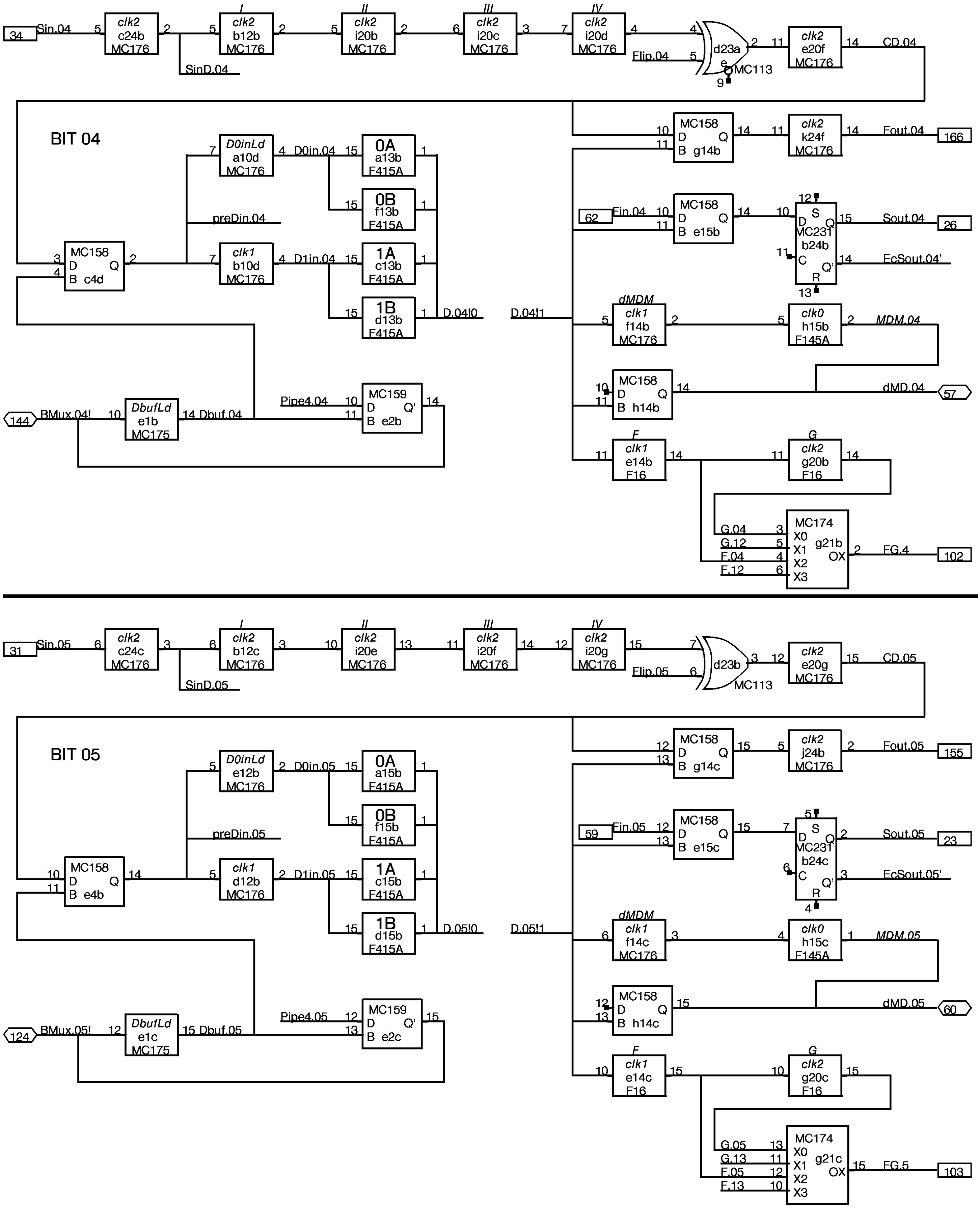
Bit 00 is on page 2



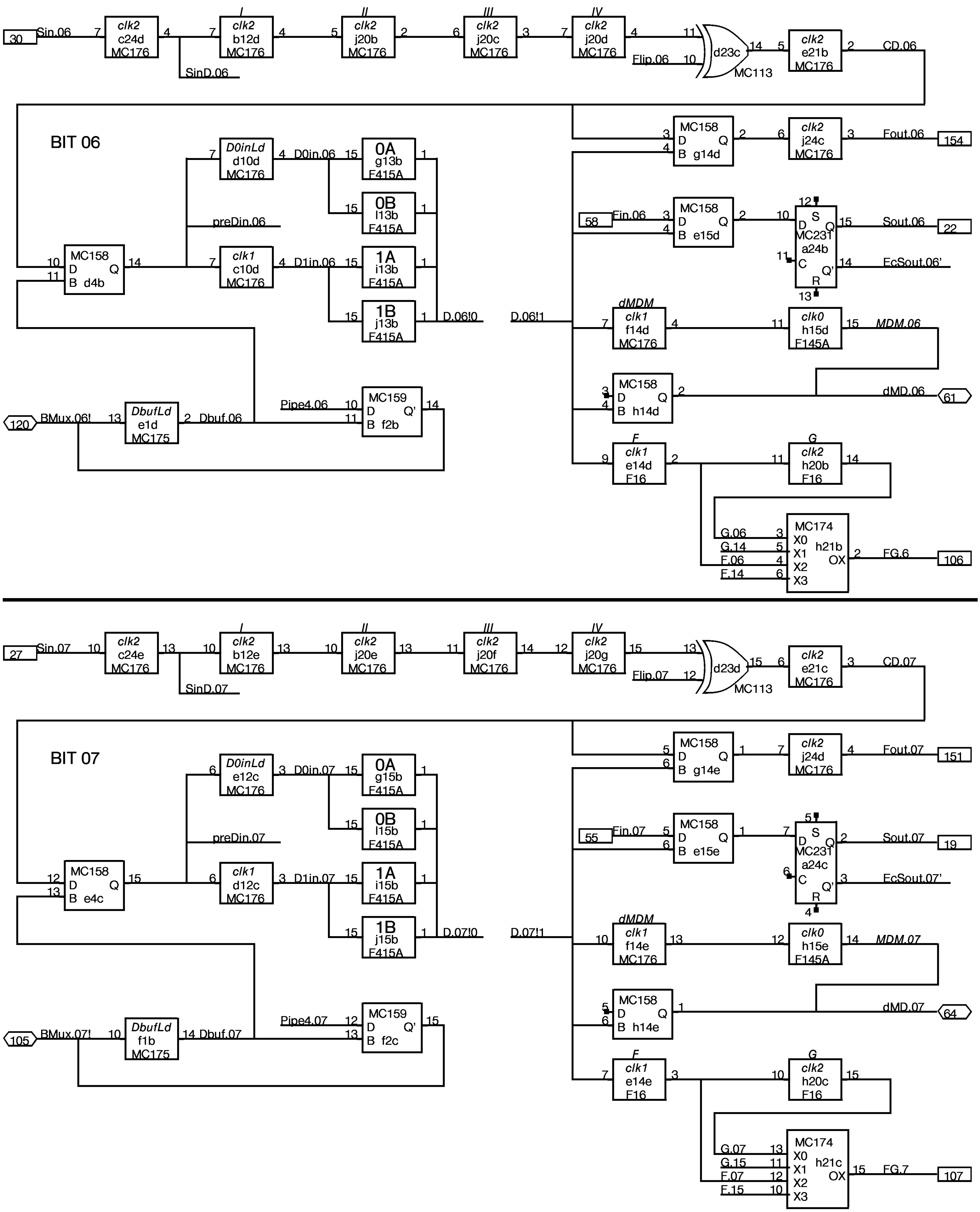
XEROX PARC/CSL	Project: Dorado	Memory data board Bit slice 01	File: MemD07.sil	Designer: D. Clark	Rev: Da	Date: 7/09/79	Page: 07



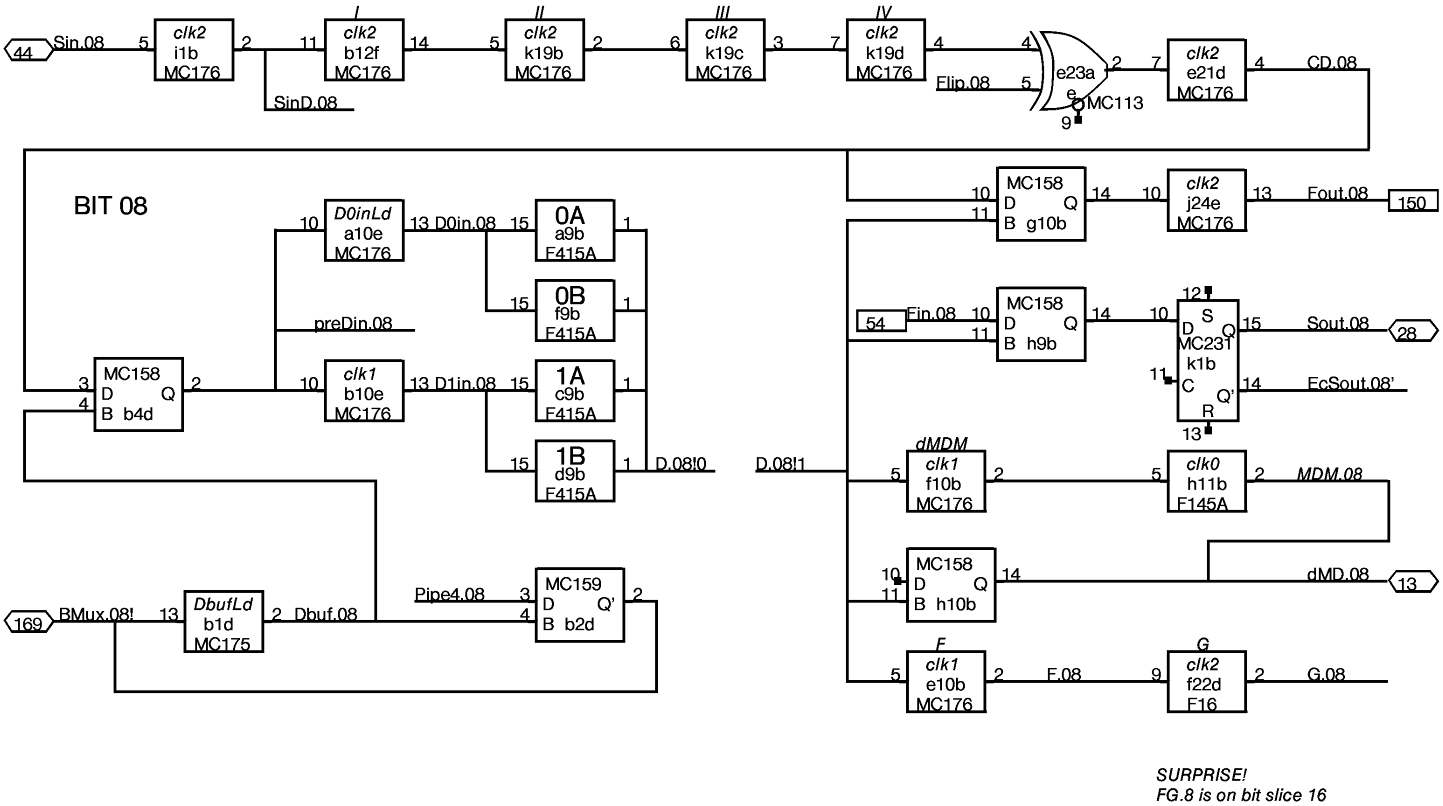
XEROX PARC/CSL	*Project* Dorado	Memory data board Bit slices 02-03	*File* MemD08.sil	*Designer* D. Clark	*Rev* Da	*Date* 7/09/79	*Page* 08

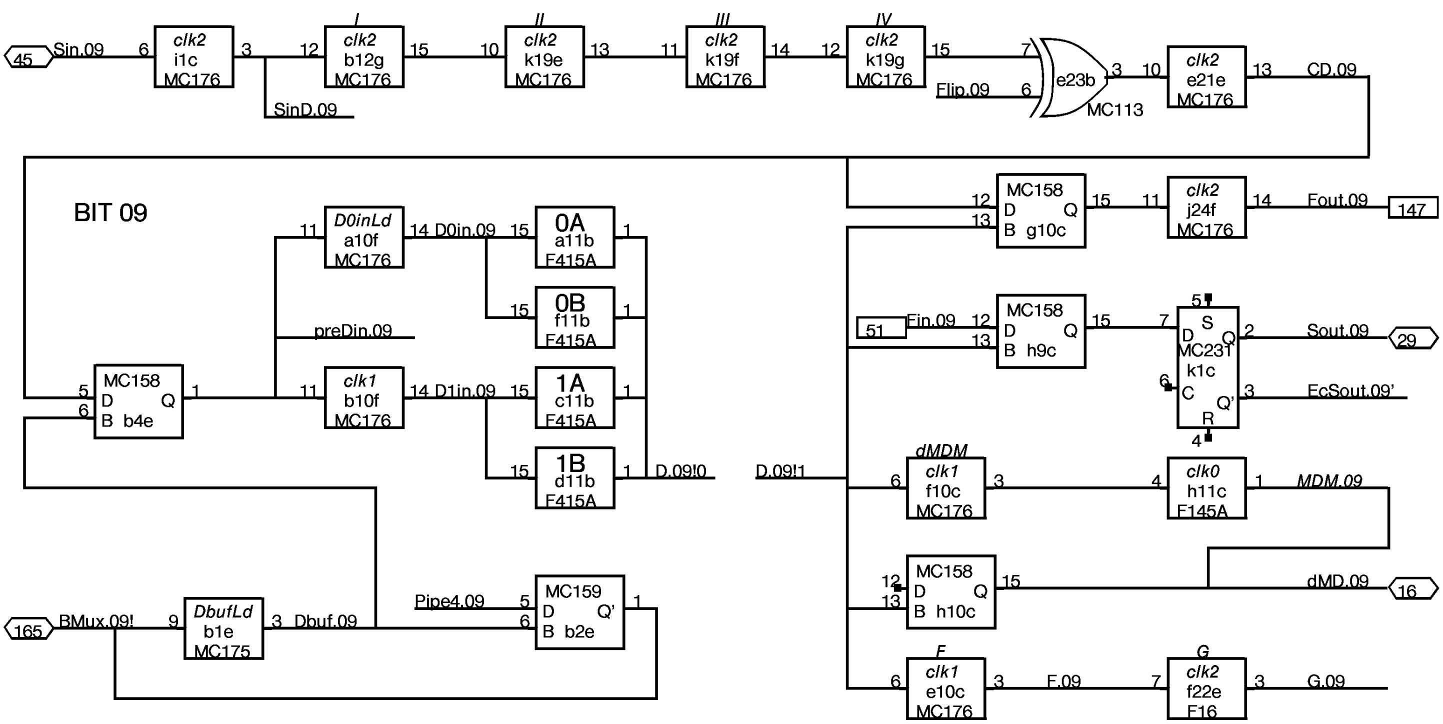


XEROX PARC/CSL	Project Dorado	Memory data board Bit slices 04-05	File MemD09.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 09

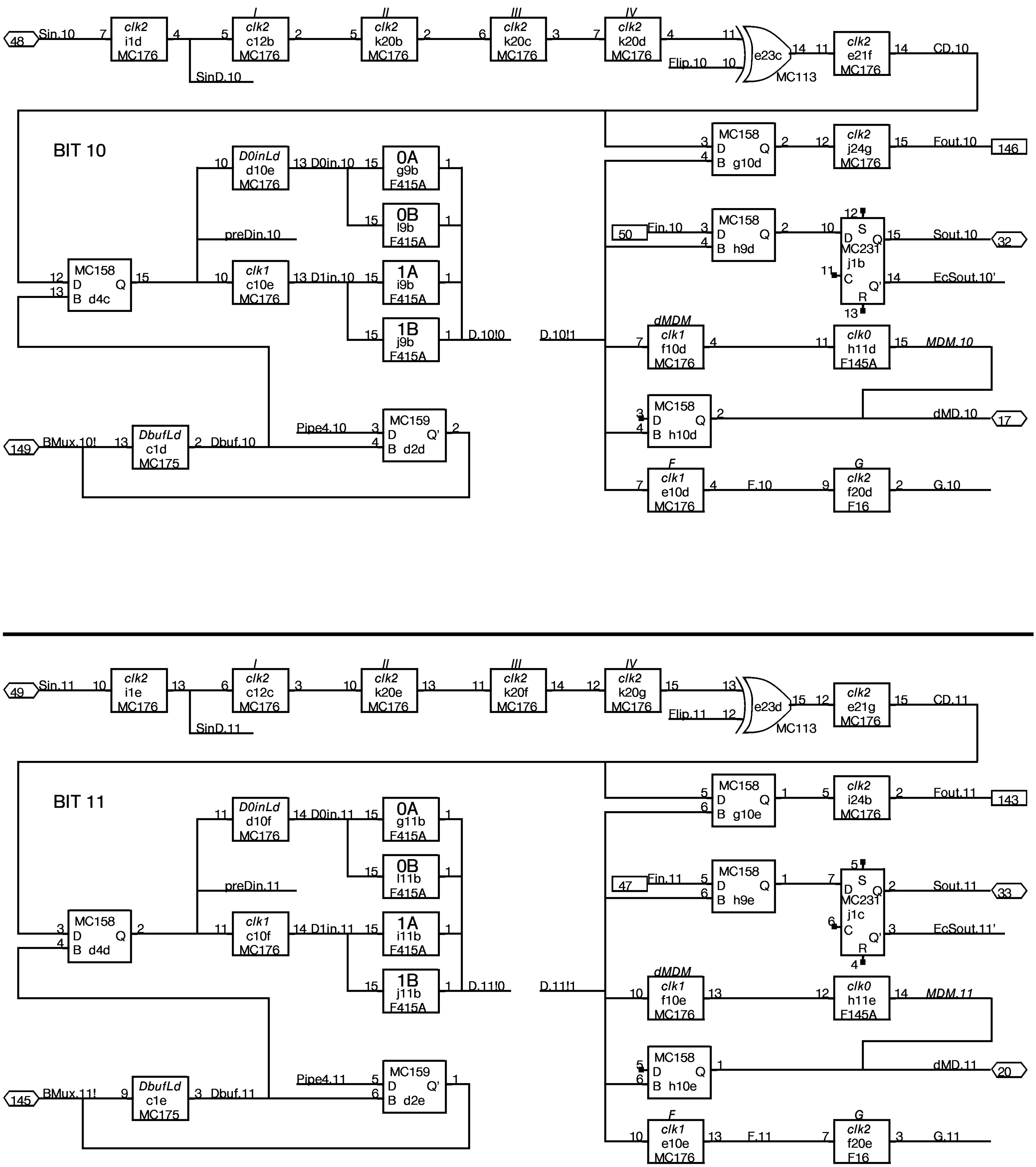


XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Bit slices 06-07	MemD10.sil	D. Clark	Da	7/09/79	10

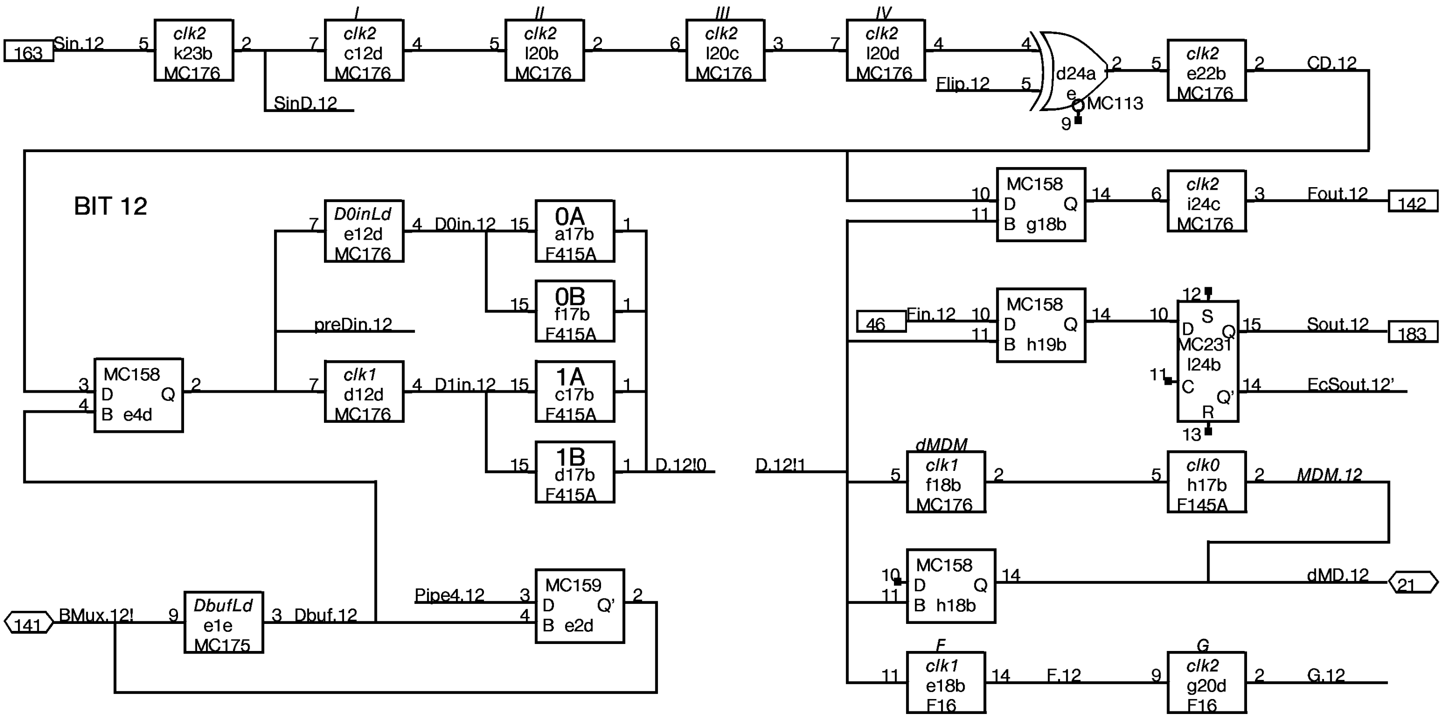


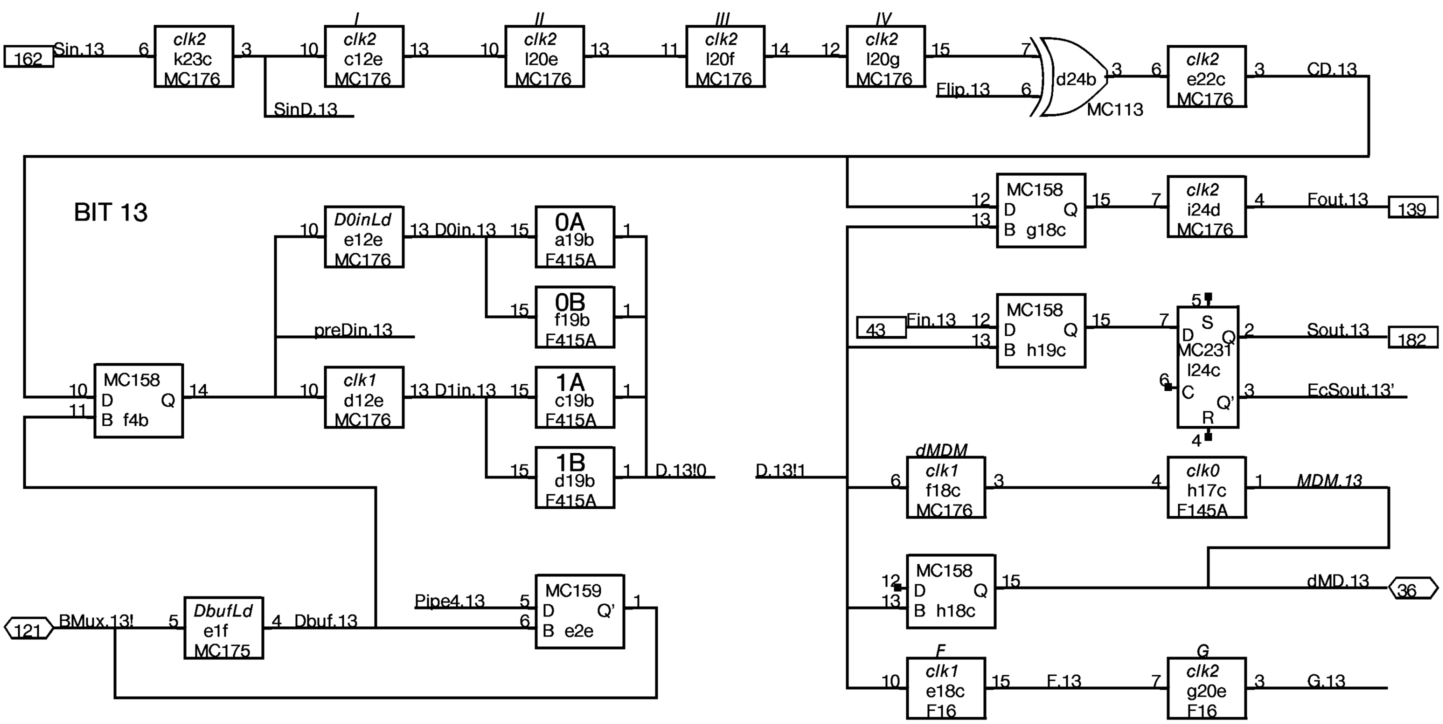


XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Bit slices 8-9	MemD11.sil	D. Clark	Da	7/09/79	11

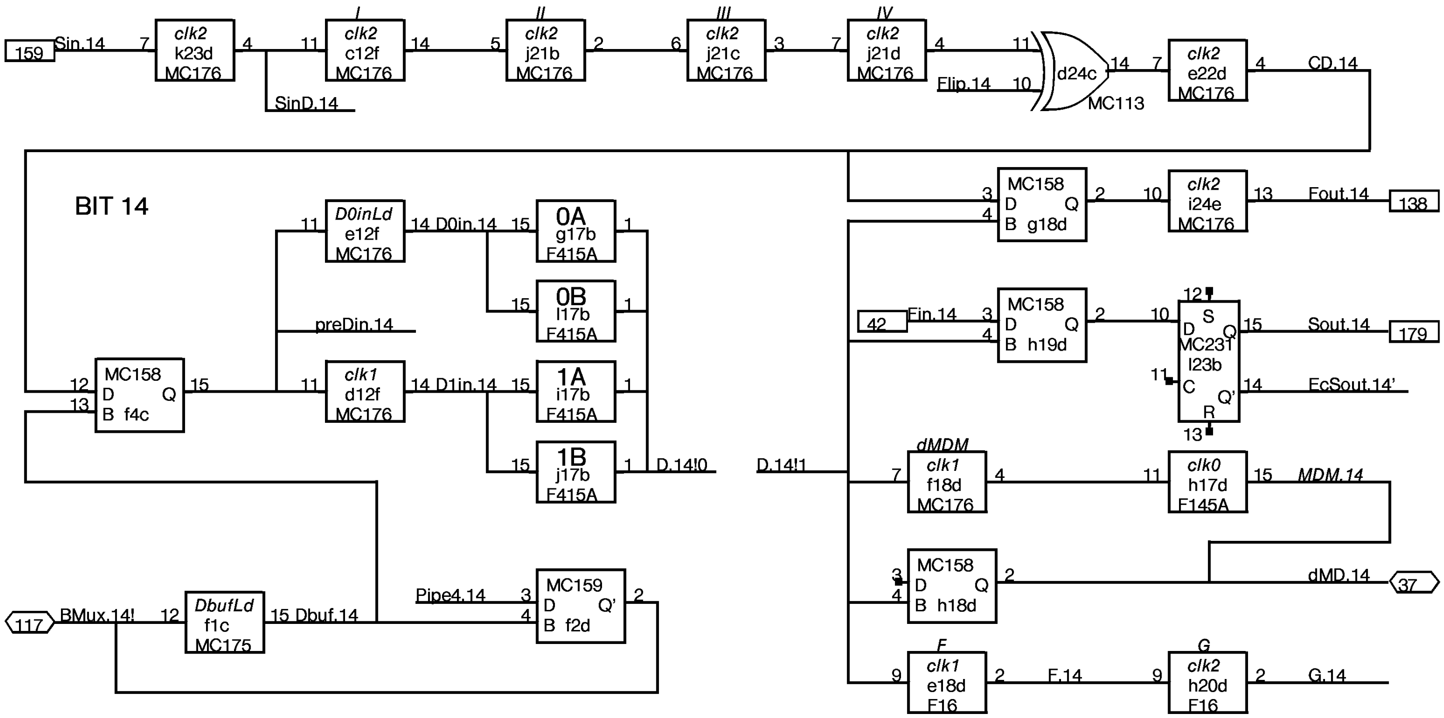


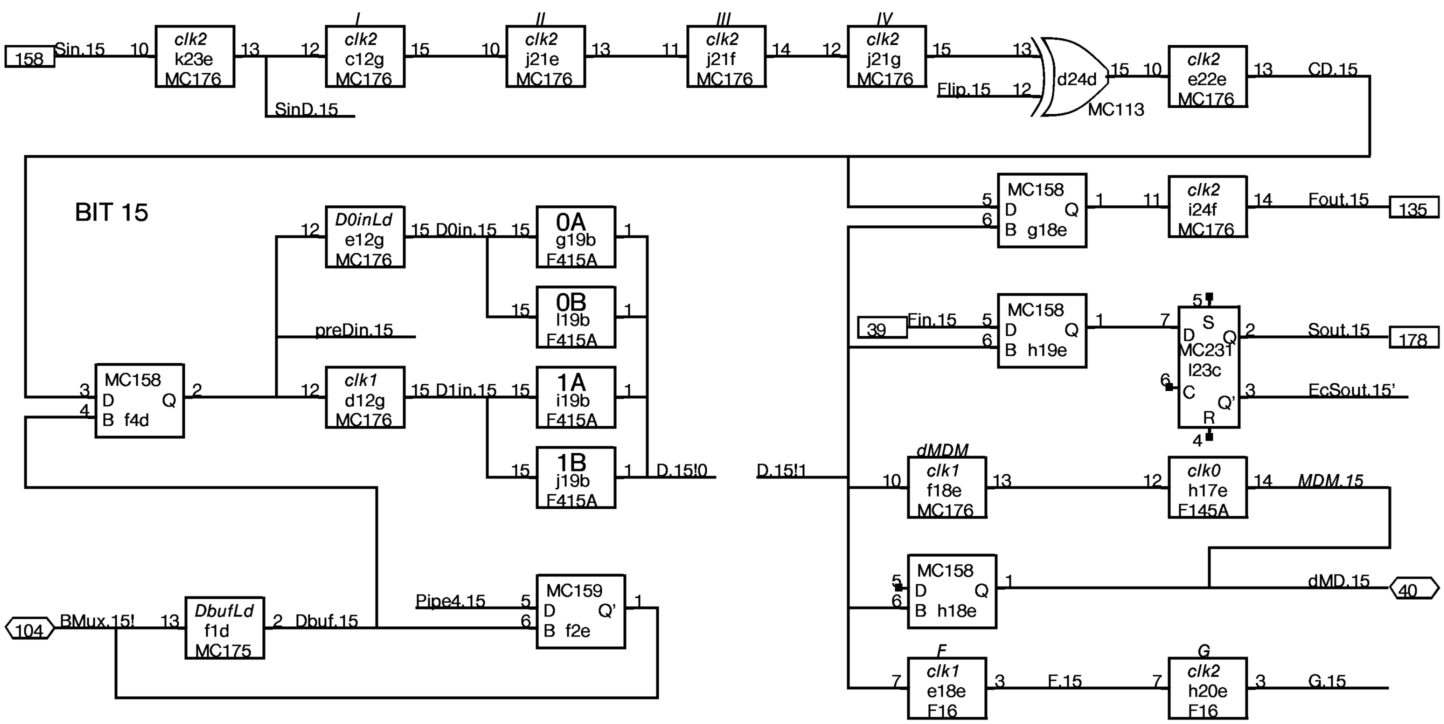
XEROX PARC/CSL	Project Dorado	Memory data board Bit slices 10-11	File MemD12.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 12



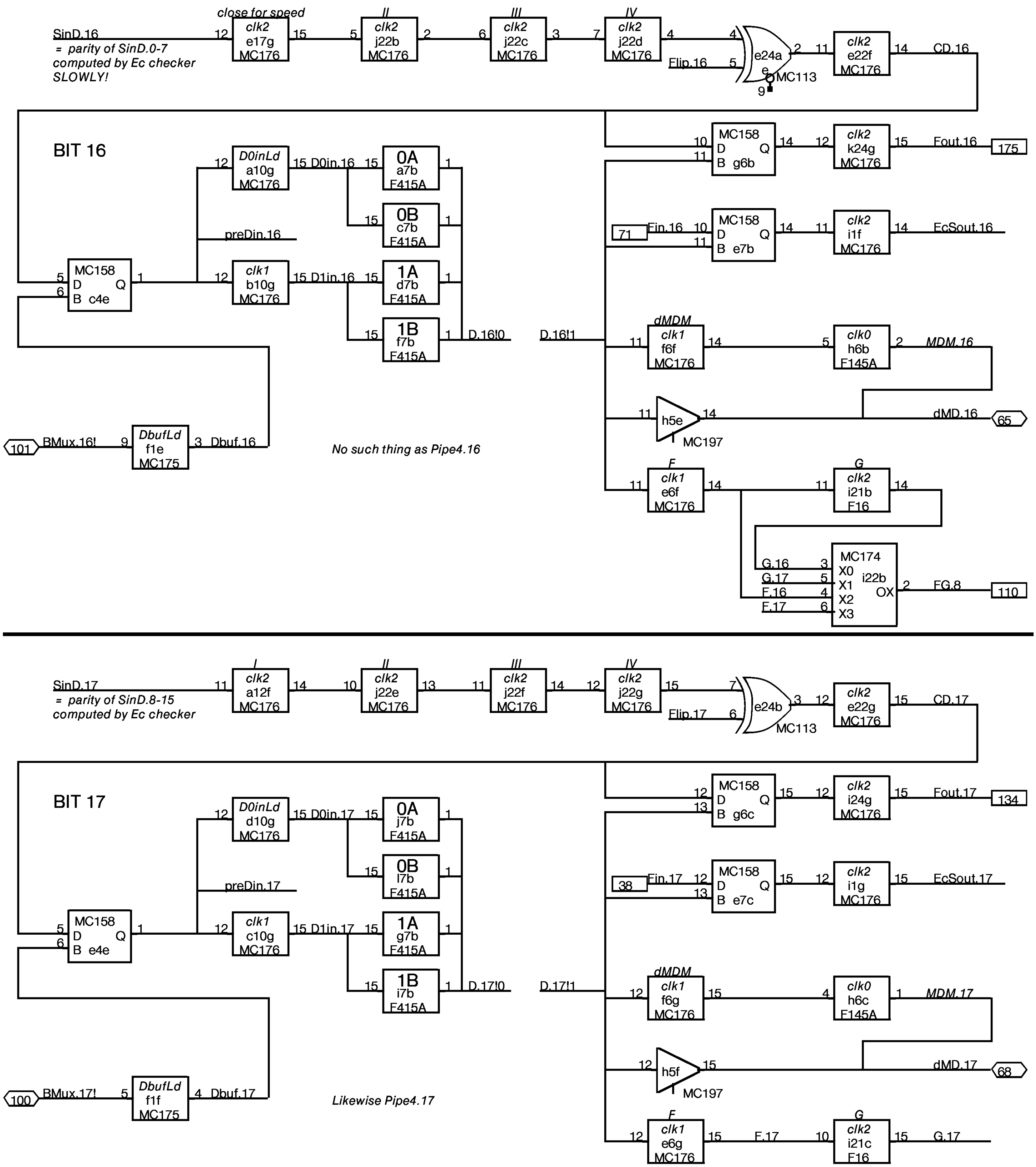


XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Bit slices 12-13	MemD13.sil	D. Clark	Da	7/09/79	13

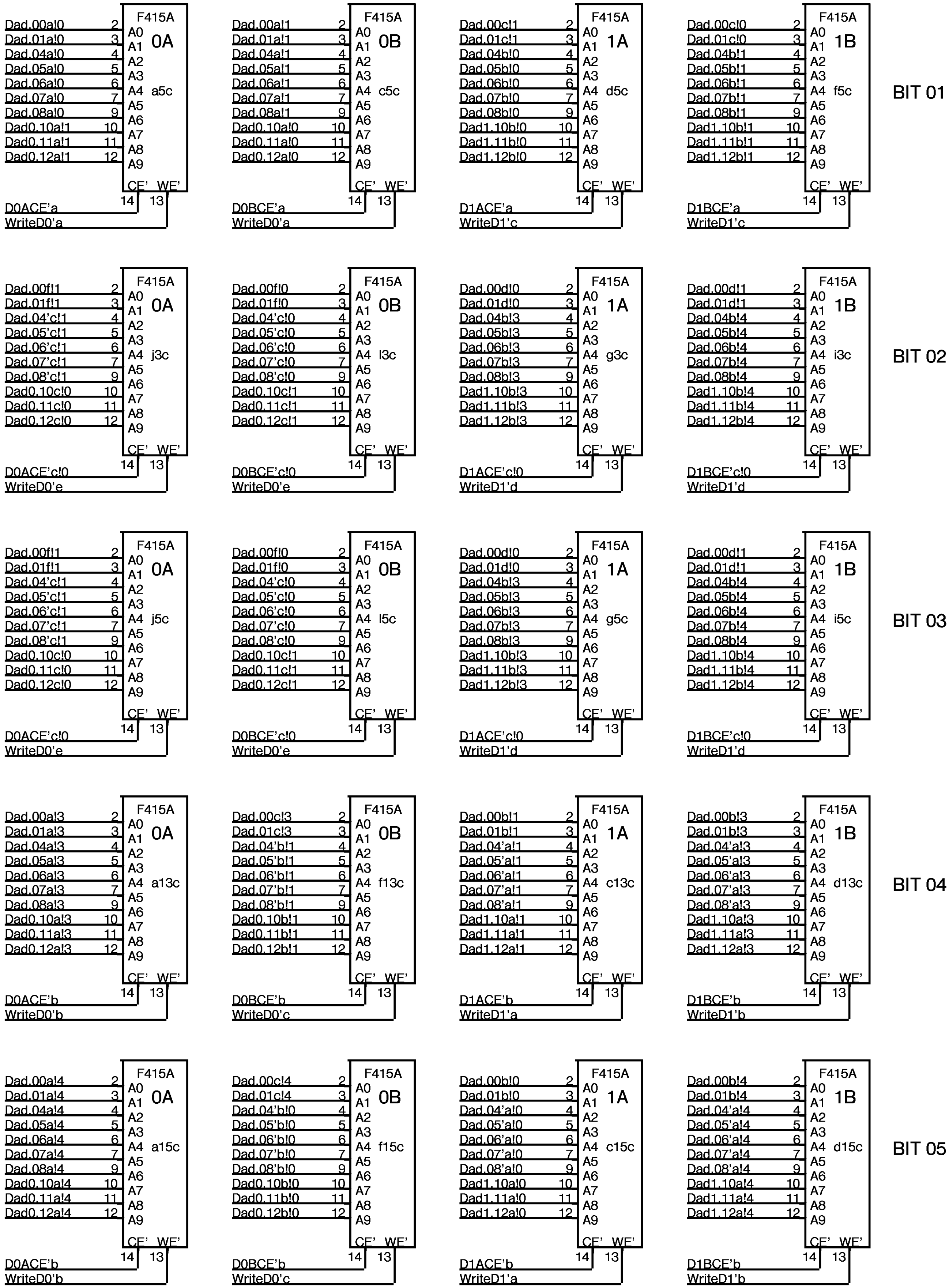




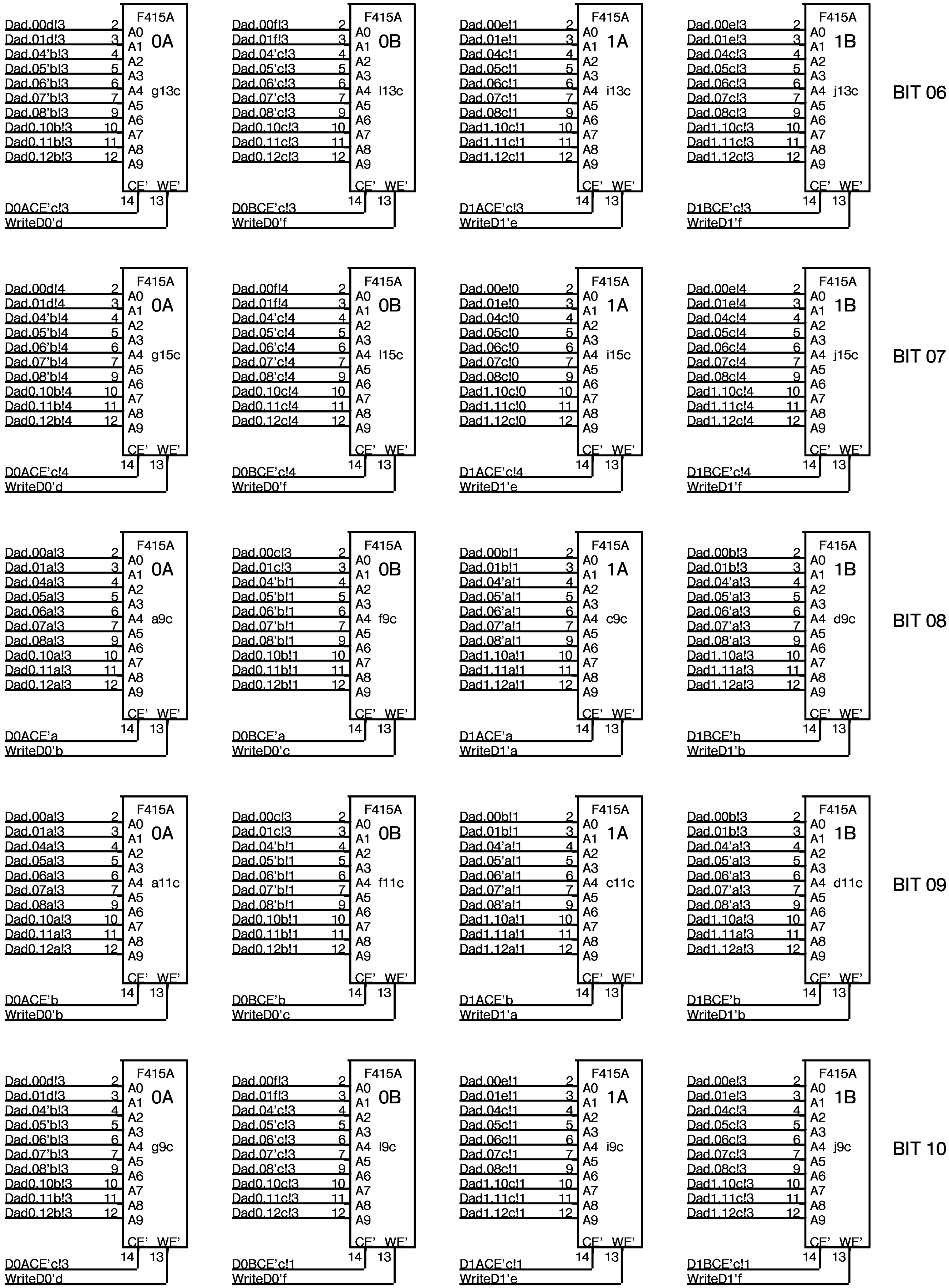
XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory Data board Bit slices 14-15	MemD14.sil	D. Clark	Da	7/09/79	14



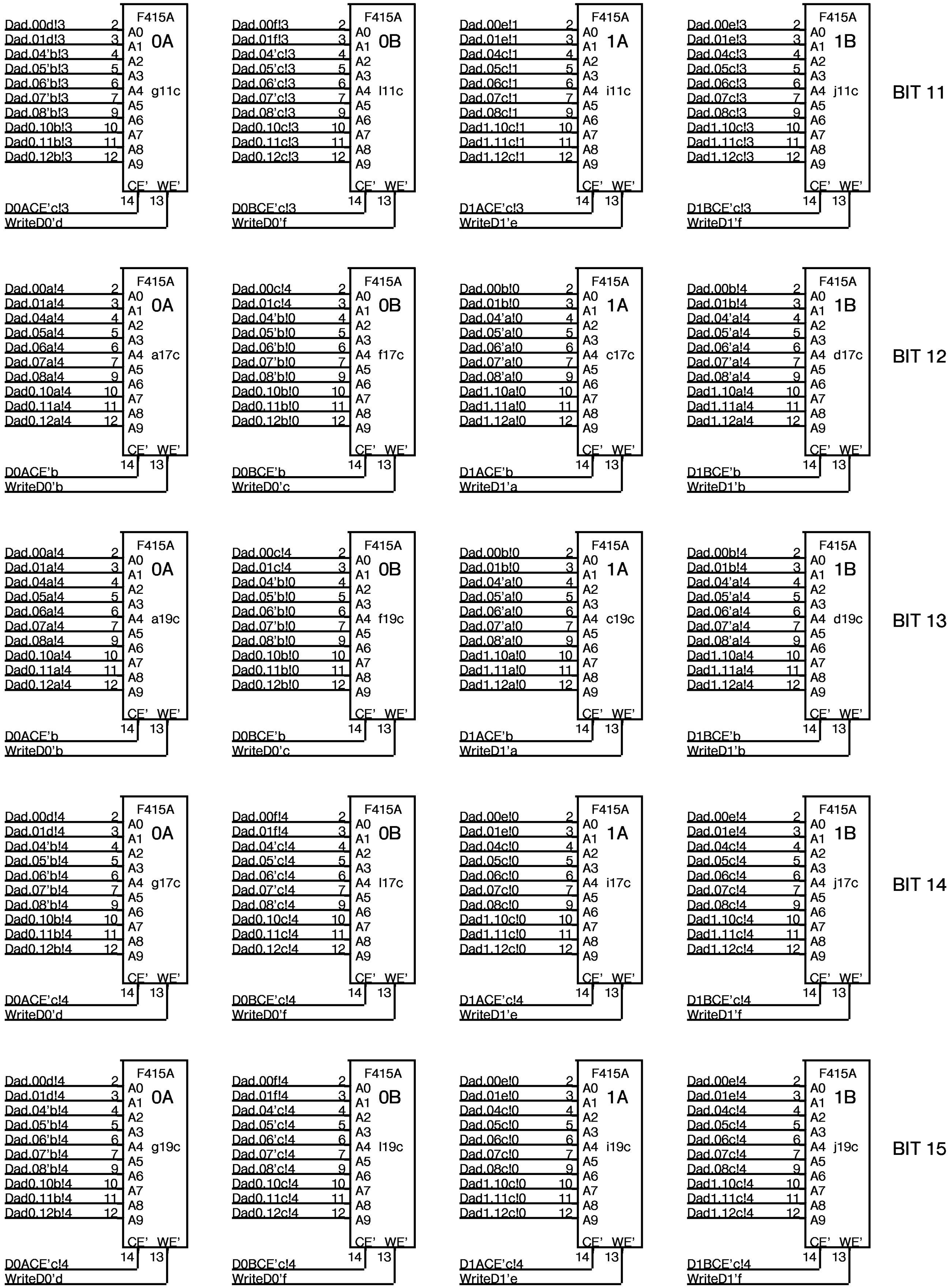
XEROX PARC/CSL	Project Dorado	Memory data board Bit slices 16-17	File MemD15.sil	Designer D. Clark	Rev Da	Date 7/10/79	Page 15



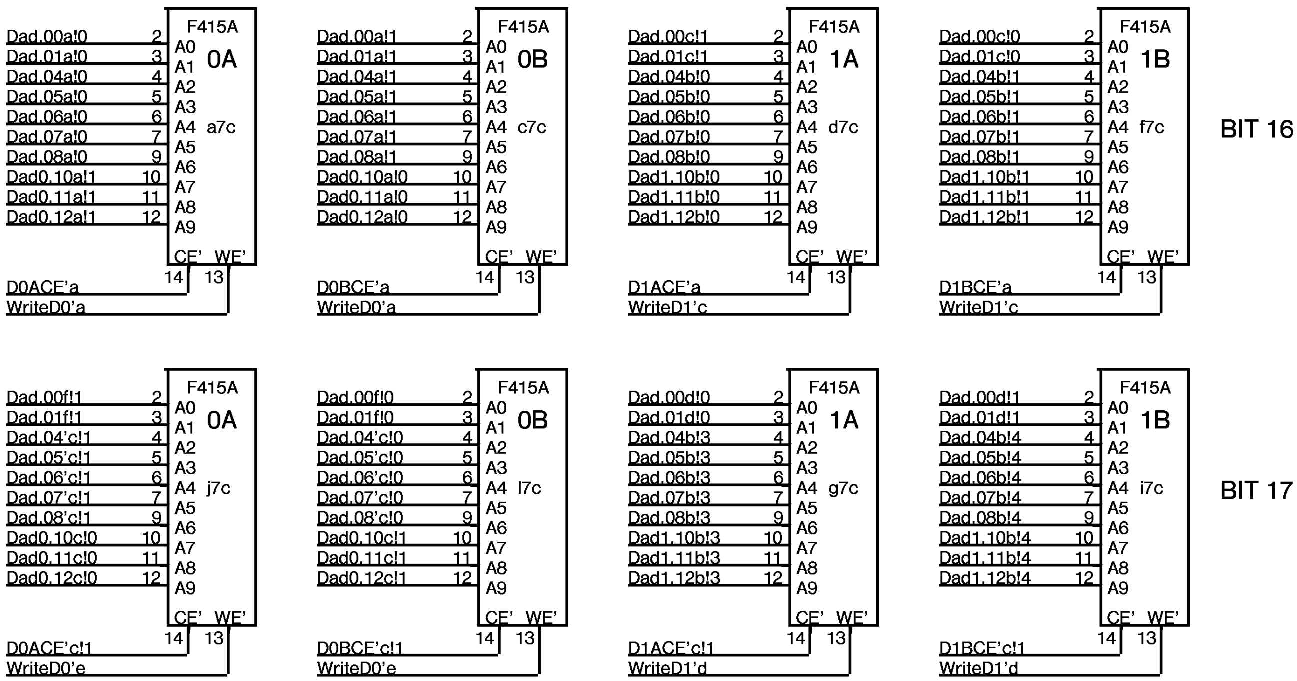
XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board D memory addressing 01-05	MemD16.sil	D. Clark	Da	7/09/79	16



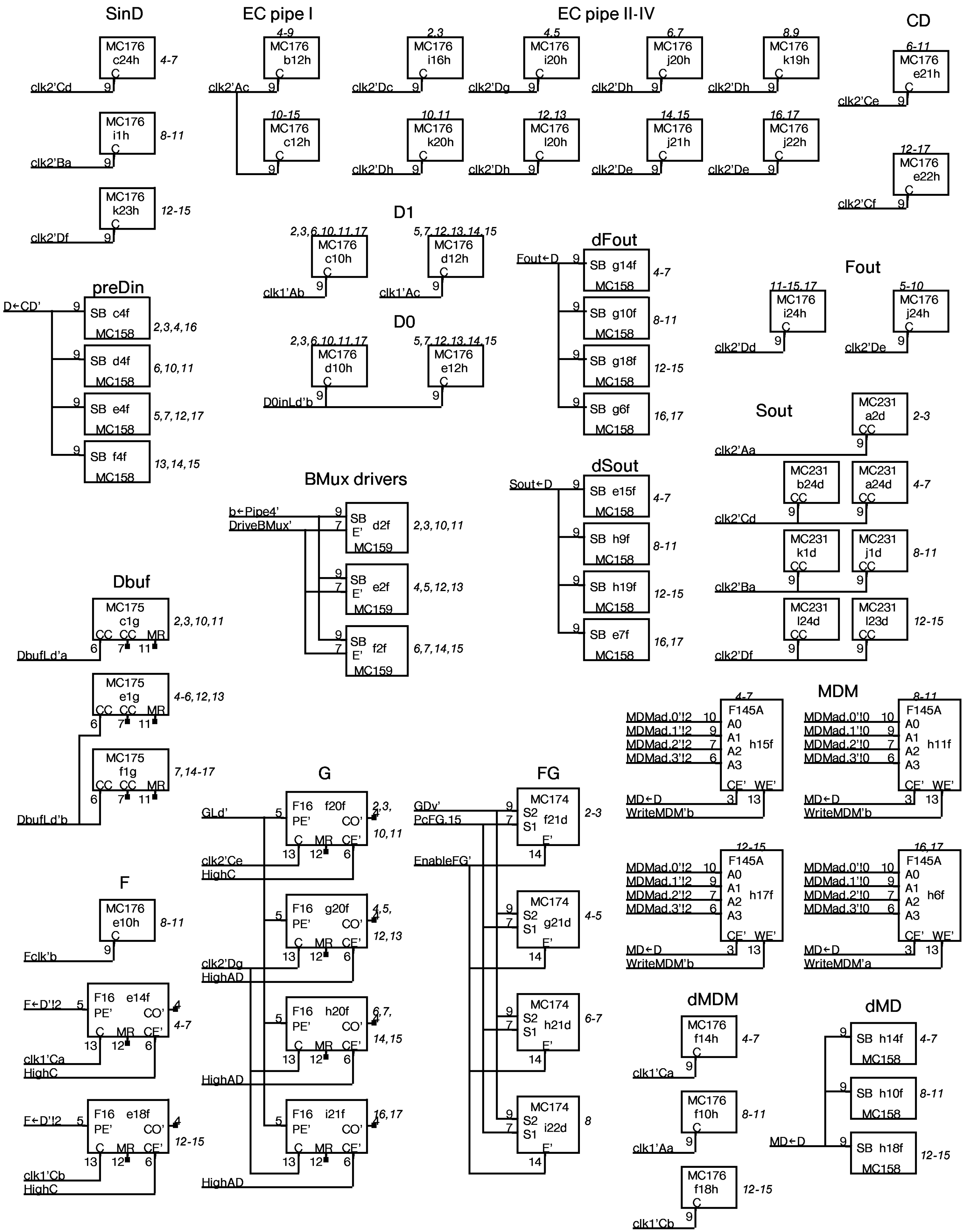
XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board D memory addressing 06-10	MemD17.sil	D. Clark	Da	7/09/79	17



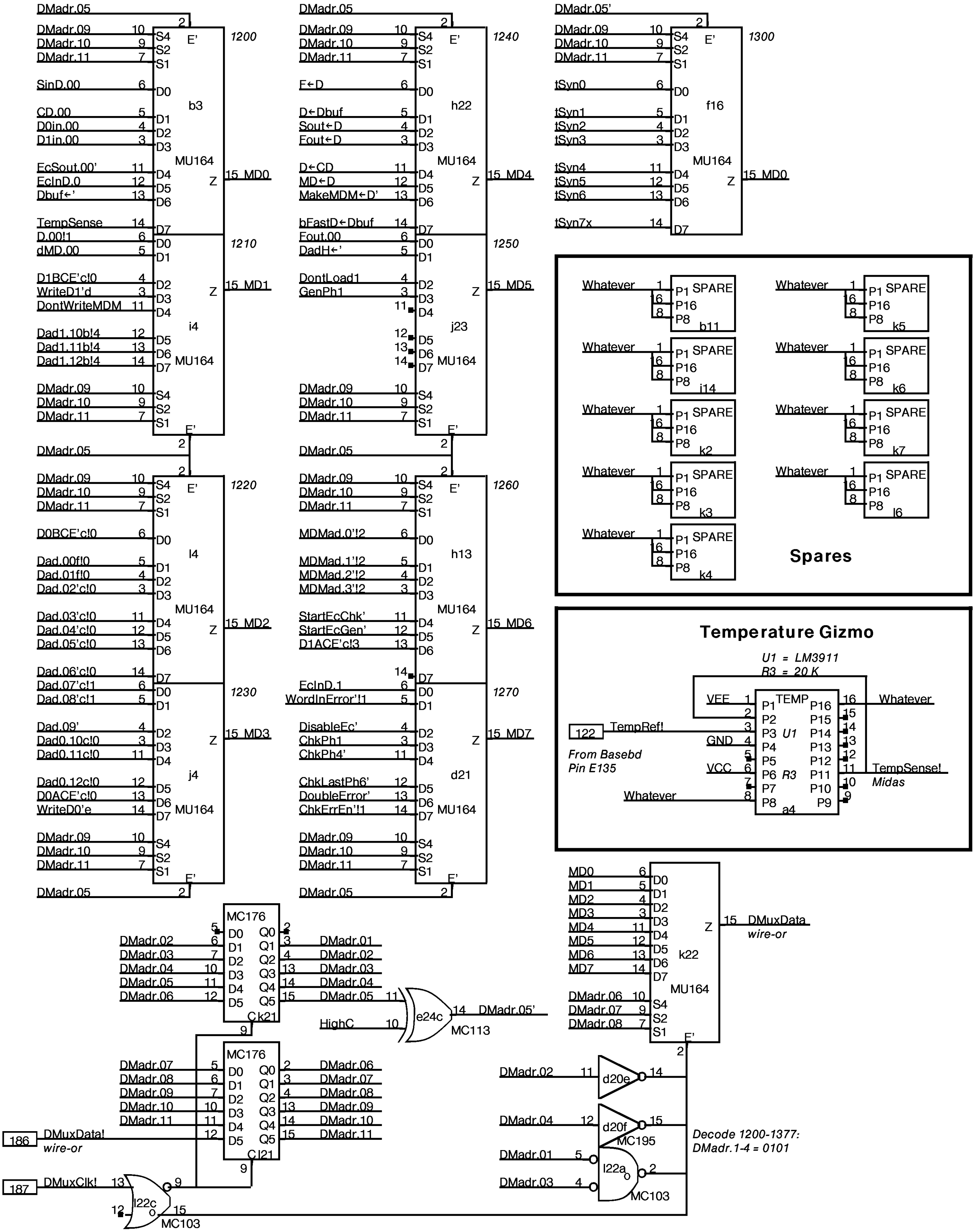
XEROX PARC/CSL	Project Dorado	Memory data board D memory addressing 11-15	File MemD18.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 18



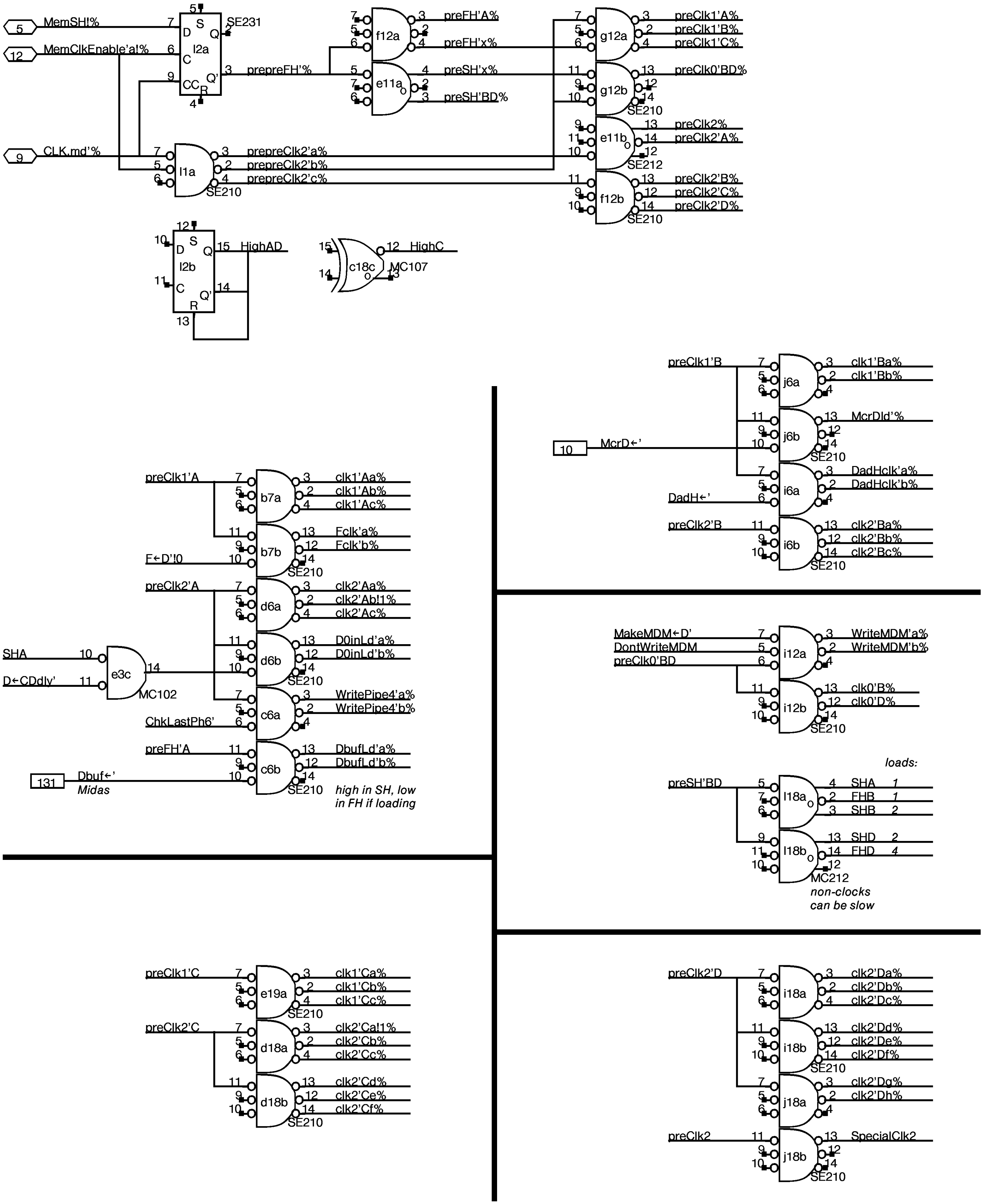
XEROX PARC/CSL	Project Dorado	Memory data board D memory addressing 16-17	File MemD19.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 19



XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Bit slice control	MemD20.sil	D. Clark	Da	7/09/79	20



XEROX PARC/CSL	Project		File	Designer	Rev	Date	Page
	Dorado	Memory data board Mufflers, Temp, Spares	MemD21.sil	D. Clark	Da	7/31/79	21



XEROX PARC/CSL	Project Dorado	Memory data board Clocks	File MemD22.sil	Designer D. Clark	Rev Da	Date 7/09/79	Page 22

Cxxx

A	a 181	b 168	c 153	d 137	e 124	f 109	g 93	h 80	i 64	j 48	k 33	l 20	B
1	Sout drvr 0-1 231 2a	BMux rcvr 0,1,8,9 175 *a	BMux rcvr 2,3,10,11 175 *a	EcOut 0,2,4,6 dSTPerr 176 2a	BMux rcvr 4-6,12,13 175 *b	BMux rcvr 7,14-17 175 *b	Dad rcvrs 2,3 231 *a	6,7 231 *b	SinD.8-11 EcSt.16,17 176 2a	Sout drvr 10-11 231 2a	Sout drvr 8-9 231 2a	Clock 210	1
2	Sout drvr 2-3 231 2a	BMux rcvr 0,1,8,9 159	SinD.0-3 EcInD.0 176 2a	BMux rcvr 2,3,10,11 159	BMux rcvr 4,5,12,13 159	BMux rcvr 6,7,14,15 159	4,5 231 *a	8,9 231 *b	0-5 TestSyn 176 *	6-8 176 *		231	2
3	D0A.00 a,a,a,a,a 415A	SinD.0 CD.0 Dbuf← D0← 1in 0 EcSout.0 EcInD.0	D0B.00 a,a,a,a,a 415A	D1A.00 c,b,b,a,c 415A	MemError ECFault preD0inLd DriveBMux 102	D1B.00 c,b,b,a,c 415A	D1A.02 d,b,b,c,d 415A	Pipe34Ad DntWrtMDM 176 0	D1B.02 d,b,b,c,d 415A	D0A.02 f,'c,c,c,e 415A		D0B.02 f,'c,c,c,e 415A	3
4	TempSense Gizmo PLAT	0,1,8,9 158	2,3,4,16 158	preDin 6,10,11 158	5,7,12,17 158	13,14,15 158	dFout 0-3 158	MdM 0-3 145 *a	CE1B D.0 10-12(1) Write1 DWrMDM dMD.0	Dad.7-9 CE0A, 10-12(0) Write0		Dad.0-6 CE0B MU164	4
5	D0A.01 a,a,a,a,a 415A	12-15 145	D0B.01 a,a,a,a,a 415A	D1A.01 c,b,b,a,c 415A	dSout 0-3 158	D1B.01 c,b,b,a,c 415A	D1A.03 d,b,b,c,d 415A	dMd 0-3,16,17 197	D1B.03 d,b,b,c,d 415A	D0A.03 f,'c,c,c,e 415A		D0B.03 f,'c,c,c,e 415A	5
6	4-7 Pipe4 145	8-11 145	clks 210	clks 210	F 0-3,16,17 176 *a	dMdM 0-3,16,17 176 1a	dFout 16,17 158	MdM 16,17 145 *a	clks 210	clks 210			6
7	D0A.16 a,a,a,a,a 415A	clks 210	D0B.16 a,a,a,a,a 415A	D1A.16 c,b,b,a,c 415A	dSout 16,17 158	D1B.16 c,b,b,a,c 415A	D1A.17 d,b,b,c,d 415A	Dad1 10-12 b F16 1a	D1B.17 d,b,b,c,d 415A	D0A.17 f,'c,c,c,e 415A		D0B.17 f,'c,c,c,e 415A	7
8	Dad0 10-12 a F16 2b0	Dad.2-5 a, 'a 101	Dad.6-8 a, 'a F←D' 101	Dad.0-1 a,b,c 210	Writes a,b,c 211	Dad.2-5 b, 'b 101	Dad.6-8 b, 'b b←Pipe4' 101	Writes d,e,f 210	Dad.0-1 d,e,f 210	Dad.2-5 c, 'c 101	Dad.6-8 c, 'c Dad.13' 101	Dad0 13 10-12 c F16 2b	8
9	D0A.08 a,a,a,a,b 415A	Dad1 10-12 a F16 1b	D1A.08 b,'a,a,a,a 415A	D1B.08 b,'a,a,b,b 415A	Dad0 10-12 b F16 2b2	D0B.08 c,'b,b,a,c 415A	D0A.10 d,'b,b,c,d 415A	dSout 8-11 158	D1A.10 e,c,c,c,e 415A	D1B.10 e,c,c,c,f 415A	Dad1 10-12 c F16 1b	D0B.10 f,'c,c,c,f 415A	9
10	D0in 0,1,4, 8,9,16 176 *a	D1in 0,1,4, 8,9,16 176 1b	D1in 2,3,6, 10,11,17 176 1b	D0in 2,3,6, 10,11,17 176 *b	F 8-11 176 *b	dMdM 8-11 ClearWA 176 1a	dFout 8-11 158	dMd 8-11 158	CE's 0A, 1A 210	CE's 0B, 1B 210	preCE 159	preCE DntLoad0 1664	10
11	D0A.09 a,a,a,b,b 415A		D1A.09 b,'a,a,b,a 415A	D1B.09 b,'a,a,b,a 415A	preclks 212	D0B.09 c,'b,b,b,c 415A	D0A.11 d,'b,b,c,d 415A	MdM 8-11 145 *b	D1A.11 e,c,c,c,e 415A	D1B.11 e,c,c,c,f 415A	B 180	D0B.11 f,'c,c,c,f 415A	11
12	0-3,17 176 2c	Ec pipe I 4-9 176 2c	10-15 176 2c	D1in 5,7,12 13,14,15 176 1c	D0in 5,7,12 13,14,15 176 *b	preclks 210	preclks 210	MDMad 141 2c	WriteMDM clk0'B,D 210	163	Generator A 180	163	12
13	D0A.04 a,a,a,b,b 415A	Cphi 172	D1A.04 b,'a,a,b,a 415A	D1B.04 b,'a,a,b,a 415A	ChkPhi F00 2a0	D0B.04 c,'b,b,b,c 415A	D0A.06 d,'b,b,c,d 415A	MDMad(4) CE1A StrtEcChk StrtEcGen	D1A.06 e,c,c,c,e 415A	D1B.06 e,c,c,c,f 415A	176 2b	D0B.06 f,'c,c,c,f 415A	13
14	0-7 163	8-15 163	SinD.17 170	ChkQWi ChkWi F16 2a0	F 4-7 F16 1a	dMDM.4-7 LastQWi 176 1a	dFout 4-7 158	dMd 4-7 158		C 180	D 180	E 180	14
15	D0A.05 a,a,a,b,b 415A	A 180	D1A.05 b,'a,a,b,a 415A	D1B.05 b,'a,a,b,a 415A	dSout 4-7 158	D0B.05 c,'b,b,b,c 415A	D0A.07 d,'b,b,c,d 415A	MdM 4-7 145 *b	D1A.07 e,c,c,c,e 415A	D1B.07 e,c,c,c,f 415A	F 180	D0B.07 f,'c,c,c,f 415A	15
16	B 180	Checker C 180	D 180	A-C 158	D-E 158	tSyni MU164	PipeII-IV 0,1 176 2c	Chk/Gen 3 dClearWA 104	PipeII-IV 2,3 176 2c	(D) D←CDdly 231 2b	A-C 158	(E) 231 2a	16
17	D0A.12 a,a,a,b,b 415A	E 180	D1A.12 b,'a,a,b,a 415A	D1B.12 b,'a,a,b,b 415A	ChkPh,etc. SinD.16dly 176 2a2	D0B.12 c,'b,b,b,c 415A	D0A.14 d,'b,b,c,d 415A	MdM 12-15 145 *b	D1A.14 e,c,c,c,e 415A	D1B.14 e,c,c,c,f 415A	D-E 158	D0B.14 f,'c,c,c,f 415A	17
18	tSyn0-3 F00 2b	tSyn4,7 231 2b	SinD.16 D←CD' HighC 107	clks 210	F 12-15 F16 1b	dMdM12-15 TickTockCE ChkLstPh6 176 1b	dFout 12-15 158	dMd 12-15 158	clks 210	clks 210	GenPhi F00 2a	FH, SH 212	18
19	D0A.13 a,a,a,b,b 415A	tSyn5,6 231 2b	D1A.13 b,'a,a,b,a 415A	D1B.13 b,'a,a,b,b 415A	clks 210	D0B.13 c,'b,b,b,c 415A	D0A.15 d,'b,b,c,d 415A	dSout 12-15 158	D1A.15 e,c,c,c,e 415A	D1B.15 e,c,c,c,f 415A	PipeII-IV 8,9 176 2h	D0B.15 f,'c,c,c,f 415A	19
20	Syni F16 2c	Corrector F00 2c	WordInError F00 2c	WdInError ChkPh,etc. DMadr' 195	CD 0-5 176 2e	G 2,3,10,11 F16 2e	G 4,5,12,13 F16 2g	G 6,7,14,15 F16 2g	PipeII-IV 4,5 176 2g	PipeII-IV 6,7 176 2h	PipeII-IV 10,11 176 2h	PipeII-IV 12,13 176 2h	20
21	DblError 170	Error ChkErrEn' 109	Syn3 D←Dbufdly 231 2c	EcCheck stuff MU164	6-11 176 2e	FG 2,3 174	FG 4,5 174	FG 6,7 174	G 16,17 F16 2g	PipeII-IV 14,15 176 2e	1-5 Midas 176	6-11 176	21
22	Flips16,17 ChkErrEn' 102	Flips 0-7 162	Flips 8-15 162	dCD 0-3 113	12-17 176 2f	G 0,1,8,9 F16 2f	preWrite 231 *	X←Ys MU164	FG 8 174	PipeII-IV 16,17 176 2e	DMuxData 164	Midas 2 Bank0CE 103	22
23	4-7 F16 1c	dPipe4 8-11 F16 1c	12-15 F16 1c	4-7 113	8-11 113	FG 0,1 174	dMD←D DntLoad1 dD←Dly 106	preWrite bTransport 102	DadH←' 231 0	Fout.0 DntLd1 DadH←' GenPh1	SinD.12-15 D←CDdlydly 176 2f	Sout drvr 14-15 231 2f	23
24	Sout drvr 6-7 231 2d	Sout drvr 4-5 231 2d	SinD.4-7 EcInD.1 ChkPh5w.o. 176 2d	12-15 113	16,17 DMadr.05' 113	EcOut 1,3,5,7 D←dly' 176 2f	MakeX rcvr 176 2d	bFastD←Dbf preWrite 1664	11-15,17 176 2d	Fout drivers 5-10 176 2e	0-4,16 176 2f	Sout drvr 12-13 231 2f	24
C	a 11	b 26	c 39	d 55	e 70	f 86	g 99	h 114	i 129	j 143	k 159	l 174	D

Exxx

D chip code: Dad.0-1,2-8,10-12,CE,Write

XEROX PARC/CSL	Project	Reference	File	Designer	Rev	Date	Page
	Dorado	Memory data board Layout	MemD23.sil	D. Clark	Da	7/09/79	23